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Experience

Level AI

Jan 2022 – Present

Director of AI Research

Remote (India), HQ: Mountain View, CA

- Building LLM-powered agentic “AI Workers” for contact-center workflows - coaching, deep transcript search, analytical insights, and deep-research-style report generation - combining multi-step reasoning, parallel tool calling, efficient context handling, and rigorous automated evaluations to solve analyst workflows end-to-end.
- Built and owned an optimized LLM inference stack for high-QPS in-house Llama-style fine-tuned models, AutoQA, and Summarization using prefix caching, chunked prefilling, `torch.compile` with CUDA Graphs, and FP8 quantization; achieved up to 3x higher throughput on vLLM and SGLang with no quality loss.
- Deployed EAGLE-3-based speculative decoding for 1.4x lower latency for low-QPS models, and accelerated bi-encoder and cross-encoder inference using `torch.compile`, yielding 1.5x faster inference for our intent classification model.
- Led the org-wide adoption of this optimized LLM inference stack across 8 production NLP services (mix of high and low-QPS), improving throughput and latency for all major AI features.
- Developed a post-training pipeline (SFT → DPO) on Llama 8B, with FlashAttention-2, FSDP2, `torch.compile`, BFD packing, async distributed checkpointing and activation checkpointing on 8xH100 GPUs to train a general-purpose model for contact center tasks which surpassed internal NLP benchmarks and exhibited strong zero-shot performance.
- Built the “Voice of the Customer” solution to discover a three-level hierarchy of customer concerns and classify conversations by product/service, issue type, and theme, achieving 75%+ accuracy across customers at <200 ms latency.

Mastercard - AI Garage

Aug 2020 – Jan 2022

Data Scientist

Gurugram, India

- Developed a graph-based representation learning algorithm for transaction fraud detection, improving AUCPR by 6% over existing methods while achieving a better training-time / performance tradeoff.
- Created a memory-efficient tabular GAN architecture, MeTGAN, which reduced memory usage by $\approx 80\%$ compared to the state-of-the-art (at the time) on datasets with high-cardinality categorical columns without any drop in performance.

Education

IIT (BHU) Varanasi, India

Jul 2016 – May 2020

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

CGPA - 9.57/10.0

Projects

Speculative Decoding Algorithms for LLM Inference

Apr 2026

- Implemented EAGLE-3, PARD, Medusa-1, Draft model, N-gram, and Suffix-Decoding algorithms in a unified speculative-decoding framework for Qwen2.5-7B, with shared target-model interfaces, greedy verification, metrics, and autoregressive baselines.
- Trained learned proposers on UltraChat prompts distilled with greedy Qwen2.5-7B-Instruct on H100 GPUs, showing up to 3.2x decode speedup, 2.3x PyTorch throughput, and 1.83x vLLM serial-latency speedup over autoregressive decoding.

KV Cache Compaction & Compression for LLM Inference

Apr 2026

- Built and benchmarked learned KV-cache compaction systems - `cartridges` and `STILL`, comparing compact-memory inference ($47\times$ compression, $38\times$ prefill speedup) against full-context baseline on long-context QA workloads.
- Implemented [post-training KV-cache compression techniques](#) in `gpt-fast` for Llama-style generation, implementing Attention Sink, L2-Norm compression, SnapKV, and H2O / Heavy-Hitter eviction.

Quantization-aware Training for LLMs

Apr 2026

- Built a lightweight single-GPU QAT workflow for Qwen/Qwen3-4B on AI-MO/NuminaMath-CoT, supporting bf16 baseline SFT, QAT fine-tuning, `torch.compile`, artifact export, and vLLM-based evaluation across FP8 / INT8 / INT4 variants.
- Showed that `int8.int8` QAT matched bf16 baseline accuracy in full-run comparisons, while the corresponding PTQ W8A8 INT8 artifact lost accuracy.

CUDA and Triton Kernels for Multi-LoRA LLM Inference

Oct 2025

- Implemented CUDA and Triton kernels for Batched Gather Matrix-Vector Multiplication (BGMV) and Segmented Gather Matrix-Vector Multiplication (SGMV) to support high-throughput multi-LoRA inference on Llama-like decoder models.
- Built prefill / decode microbenchmarks and end-to-end generation benchmarks comparing loop, grouped-BMM, BGMV, and SGMV paths across mixed-adaptor batches with random per-sample routing.

LLM Post-Training with SFT + GRPO in a Verifiable Multi-turn Environment

Sept 2025

- Built an end-to-end post-training pipeline for Qwen3-4B on a multi-turn Wordle task: generated synthetic teacher traces, converted them into chat-formatted SFT / GRPO datasets, and enforced strict `<think>` / `<guess>` output formatting.
- Implemented SFT with FSDP2, FlashAttention-2, gradient checkpointing, and optional 2D parallelism (FSDP2+TP / FSDP2+CP), then ran GRPO over a strict multi-turn Wordle environment backed by a vLLM rollout server.

Sparse Matrix Computations in CUDA

Sept 2024

- Implemented CUDA / C++ kernels for sparse matrix conversion, SpMV, and SpMM across COO, CSR, CSC, BSR, BSC, and ELL formats.
- Structured the project with per-kernel build targets and libtorch-based correctness checks to deepen understanding of sparse layouts, memory access patterns, and GPU kernel design.

Open Source / Technical Writing

Annotated ML Papers | Blog

- Maintain public notes and technical writing on deep learning, NLP, and ML systems; the Annotated ML Papers repository has 279+ GitHub stars.
- Write concise deep-dives on topics such as cross-encoder inference with `torch.compile`, CUDA scan kernels, multi-head latent attention, and LLM systems internals.

ML Paper Implementations

- Curate reproducible implementations of ML systems and deep learning ideas spanning FlashAttention, LLM sampling, KV cache compression, tokenization, and attention variants.
- Use the collection as a public sandbox for turning research papers into readable, runnable code and experimentation notes.

Publications

- **MeTGAN: Memory Efficient Tabular GAN for High Cardinality Categorical Datasets** - *Shreyansh Singh, Kanishka Kayathwal, Hardik Wadhwa and Gaurav Dhama* at the 28th International Conference on Neural Information Processing (ICONIP), 2021
- **CuRL: Coupled Representation Learning of Cards and Merchants to Detect Transaction Frauds** - *Maitrey Gramopadhye*, Shreyansh Singh*, Kushagra Agarwal, Nitish Srivasatava, Alok Singh, Siddhartha Asthana and Ankur Arora* at the 30th International Conference on Artificial Neural Networks (ICANN), 2021 (* $\equiv$ Equal contribution)
- **IIT (BHU) Varanasi at MSR-SRST 2018: A Language Model Based Approach for Natural Language Generation** - *Shreyansh Singh, Avi Chawla, Ayush Sharma and A.K. Singh* in Proceedings of the 1st Workshop on Multilingual Surface Realisation at the 56th Association for Computational Linguistics (ACL), 2018

Technical Skills

Languages: Python, C/C++, CUDA, Triton, SQL, Bash

ML / LLM Systems: PyTorch, Transformers, TRL, FSDP2, FlashAttention, `torch.compile`, CUDA Graphs, vLLM, SGLang, TensorRT-LLM

Core Areas: LLM post-training (SFT, DPO, GRPO), inference optimization, quantization, distributed training, CUDA / Triton kernel development, agentic systems, retrieval / ranking

Infra / Tools: Kubernetes, Linux, Git

Achievements

- Granted one provisional US patent for my work on Voice of the Customer and AgentGPT at Level AI.
- Granted two US patents for my work at Mastercard: [one for leveraging reinforcement learning and NLP to suggest charities based on news articles](#), and another for enhancing Mastercard's Threat Scan via synthetic fraud transaction generation using MeTGAN.
- [Earned a silver medal by ranking in the top 5% \(115th among 2426 teams\)](#) while participating solo in the Kaggle Shopee Price Match Guarantee competition, 2021.
- Ranked 55th (top 10%) in the Multi-dataset Time Series Anomaly Detection challenge, KDD Cup 2021.
- [Ranked 15th in CryptoHack CTF](#) (as of May 2020), a modern-day cryptography-focused Capture the Flag event.
- Recipient of the student scholarship to attend Black Hat Asia 2019 in Singapore, where 100 students were selected from 82 countries.